

LSI-EGT: Life Stressors Identification Using Context-Aware Embedding and Graph Transformer

Minni Jain  and Roopal Shakya 

Abstract—The increasing prevalence of stress-related issues highlights the urgent need for effective stress detection and intervention strategies in counseling. Stress, a common precursor to various psychological and physical health problems, arises from multiple sources and manifests through diverse life events. Understanding these sources or stressors is essential for providing targeted and effective counseling. Stressors and life events often influence and compound each other, creating a network of interrelated factors contributing to an individual's overall stress level. This study proposes a contextualized MentalRoBERTa and Graph Transformer based model to identify these stressors and significant stressful life events. Graph transformer allow nodes to selectively attend to messages received from neighbors using multi-head attention, enhancing their capacity to understand intricate relationships. The proposed model was evaluated on the stress annotated dataset (SAD), and a synthetically generated dataset. The results show that the proposed model outperforms existing methods, achieving F1-scores of 67.93 on the SAD dataset, and 97.26 on the SMOTE dataset. These findings demonstrate the effectiveness of our approach in accurately identifying and classifying stressors, making it a valuable tool for stress analysis and intervention.

Index Terms—Graph neural networks (GNNs), graph transformer, MentalRoBERTa mental health, stressors detection.

I. INTRODUCTION

STRESS has become increasingly common in contemporary society, with a notable surge following the COVID-19 pandemic [1]. The prolonged nature of the pandemic has increased stress levels across various demographic groups [2]. Social media usage has also been linked to heightened stress levels, contributing to the overall stress burden [3]. According to a survey by the American Psychological Association (APA), Generation Z is particularly affected [4], reporting higher levels of mental health concerns compared with other generations. Additionally, a systematic review of mental health changes in children and young people [5] before and during the pandemic reveals an increase in stress among both children and adults. Twenge et al. [6] found a sharp rise in anxiety and depression among teens correlating with increased screen time and social media use. For

working adults, job insecurity, work-life balance challenges, and economic instability are major contributors to stress. American Psychological Association [7] indicates that economic concerns, job loss and financial instability have worsened stress for minority groups, while older adults experience heightened stress from health issues and social isolation, especially during the pandemic. National Academies of Sciences, Engineering, and Medicine [8] links social isolation and loneliness in older adults to higher rates of mental health issues. Constant exposure to negative news and information [9] overload further exacerbates stress. Among women, work presents challenges with balancing long hours, potential pay inequities, and discrimination against the backdrop of household and childcare responsibilities [10]. Financial burdens, especially for single mothers or those in low-income brackets, add another layer of stress [11]. Studies reveal a connection between chronic stress during pregnancy and an increased risk of developing postpartum depression [12]. Polycystic ovary syndrome (PCOS) can have significant psychological implications including its association with depression, anxiety, and body image concerns [13]. Stress arises from various sources, including academic pressure, health problems, social or political concerns, and financial problems [14]. However, stress is not always detrimental [15]; it can be positive when it motivates people to overcome challenges or improve performance.

Numerous studies have explored the relationship between stressful life events and the risk of major depressive episodes. The Holmes and Rahe stress scale [16] and the presumptive stressful life events scale (PSLES) [17] were developed to measure the impact of various stressors. These scales quantify the stress associated with different life events and their relevance to mental health outcomes. Cochrane and Robertson [18] refined these scales to better capture the nuances of stress impacts. According to a 2022 APA survey [19], 62% of psychologists reported more referrals in 2021, with 68% of those with waitlists seeing longer delays, emphasizing the need for automated stress counseling solutions due to the mental health crisis. This article addresses the gap in research by identifying 32 common stressors and exploring their role in developing automated stress counseling systems to improve stress detection and intervention for better mental health outcomes. The proposed study has the following contributions:

- 1) The study identifies 32 of the most common psychiatric stressors and uses a dataset of 6128 instances, significantly broadening the scope of the research.

Received 1 July 2024; revised 30 March 2025 and 11 May 2025; accepted 20 May 2025. Date of publication 10 July 2025; date of current version 3 December 2025. (Corresponding author: Minni Jain.)

The authors are with the Department of Computer Science and Engineering, Delhi Technological University, New Delhi 110042, India (e-mail: minnijain@dtu.ac.in; shakyaaroopal121@gmail.com).

Digital Object Identifier 10.1109/TCSS.2025.3573155

TABLE I
COMPARATIVE ANALYSIS OF RELATED WORK

Authors	Year	Dataset Sources	Methods	Contributions	Results
Li et al. [20]	2017	Tencent Weibo	Poisson-based probability model	Identifies adolescents' stressful periods & triggering stressor events from social media posts.	F1-Score: 73.4%
Du et al. [21]	2018	Twitter	Recurrent Neural Network (RNN)	Identifies psychiatric stressors in suicide-related tweets.	F1-Score: 83%
Zhang et al. [22]	2019	CEGS N-GRID [32]	Named Entity Recognition (NER)	Classifies psychiatric stressors from clinical text & analyzes their association with suicidal behaviors.	F1-Score: 89.01%
Leung et al. [23]	2022	Reddit	Latent Dirichlet Allocation (LDA)	Analyzes social media posts to identify stressors related to COVID-19.	Topics: coronavirus, social relationships, family, educational and occupational, mental health, pandemic uncertainty.
Sawalha et al. [24]	2022	AVEC 2019 [33]	Random Forest	Identifies PTSD in individuals using transcribed speech of interview data.	F1-Score: 79.0%
Han et al. [25]	2022	MIMIC-III [34]	BERT	Extracts social determinants of health from clinical text.	Micro-F1-Score: 69.0%
Richie et al. [26]	2023	MIMIC [34] and UW clinical notes [35]	BioClinical-BERT based NER	Extract SDOH event triggers and their identify corresponding arguments.	F1-Score: 88%
Raza et al. [27]	2023	Electronic case reports (LitCOVID API) [36]	BioBERT + BiLSTM	Extracts SDOH from clinical texts.	F1-Score: 92.98%
Raveau et al. [28]	2023	Chilean-based NGO 'It Gets Better' Project - Safe Hour Program	Logistic regression	Identifies six key stressor categories and their association with mental health conditions.	Accuracy- Self-image: 73%, Violence: 68%, Sexual diversity: 93%, Relational: 80%, Emotional crisis: 66%, Performance: 71%
Iparraguirre-Villanueva et al. [29]	2024	Patient health questionnaire (PHQ) of college students	Logistic regression	Categorizes and predicts student depression.	F1-Score: 72%

- Introduces a pairwise cosine similarity-based graph construction approach and implements graph transformer that utilizes a multihead attention approach to effectively identify stressors.
- Demonstrates advancements over current state-of-the-art models in identifying psychiatric stressors from text.

The subsequent Section II explores existing research on stressor detection and identifies the gaps. Section III introduces the model architecture, including graph construction with cosine similarity and Graph Transformer. Section IV presents dataset details, results, and analysis. Section V summarizes contributions and future directions.

II. RELATED WORK

The field of stress detection using text-based datasets has seen significant advancements, with a growing focus on identifying stressors or causes of stress directly from textual data. This section reviews recent research efforts in this area, highlighting methodologies, findings, and the impact of these studies. A summary of recent works is provided in Table I.

In the realm of text-based stressor detection, Li et al. [20] examined adolescents' stressful periods and the stressor events that triggered them using a Poisson-based probability model. The findings revealed that self-cognition and school life were the primary stressor domains. Furthermore, Du et al. [21] investigated the detection of psychiatric stressors related to suicide using CNN and RNN based classifiers in Twitter data. A CNN-based classifier was used to identify suicide-related tweets. Zhang et al. [22] employed CRF-based named entity recognition (NER) with BIO tagging, utilizing features from

NLP, domain knowledge, and word embeddings for recognizing and classifying psychiatric stressors in clinical text. Leung and Khalvati [23] utilized latent Dirichlet allocation (LDA) to social media data to identify the psychosocial stressors during the COVID-19 pandemic. Sawalha et al. [24] developed a machine learning model using sentiment analysis to identify PTSD in individuals through virtual semistructured interviews. Han et al. [25] used CNN, LSTM, and BERT models to extract 13 social determinants of health (SDOH) from clinical text, with BERT outperforming other models. Richie et al. [26] proposed a two-stage architecture using BioClinical-BERT for named entity recognition, first extracting SDOH event triggers (e.g., substance use, employment) and then identifying corresponding arguments (e.g., alcohol type) through a multitask, multilabel NER model. Raza et al. [27] developed a NER model for extracting SDOH from COVID-19 case reports, achieving a 92.98% F1-score using semisupervised learning. Raveau et al. [28] analyzed mental health chat conversations from the Safe Hour helpline, findings reveal increased discussions on self-image and interpersonal issues, while performance-related topics declined. Iparraguirre-Villanueva et al. [29] used machine learning to classify student depression, finding logistic regression had the highest accuracy. Key findings indicated that two out of five students experienced mild depression, 90% of depressed students did not seek treatment, and obesity significantly increased depression risk.

Some studies have applied graph neural networks for depression detection. Li et al. [30] introduced a method to extract depression states using a heterogeneous graph attention neural network. Furthermore, Jain et al. [31] combined multiple

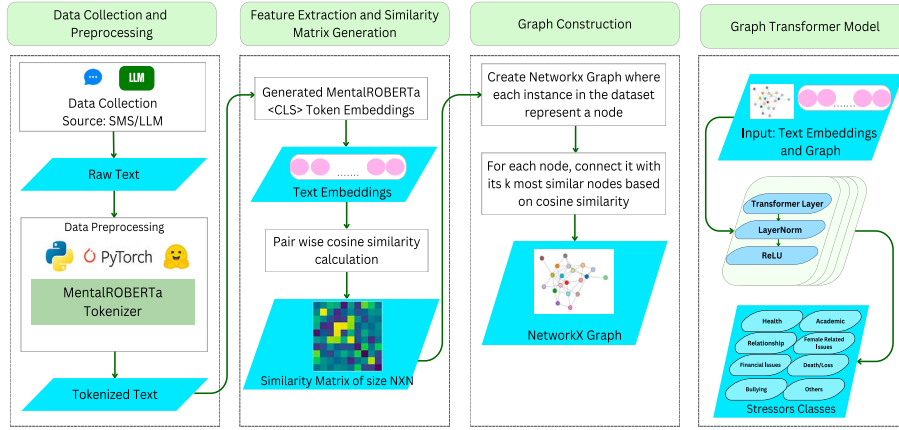


Fig. 1. Architecture of the proposed methodology (LSI-EGT).

token-level embeddings, including BERT, RoBERTa, and DeBERTa, with graph attention networks (GATs) to detect depression.

Previous research has often been limited in scope, focusing on detecting only a few stressors due to a lack of comprehensive datasets. In our study, we have identified 32 of the most common stressors among individuals. To identify these stressors, we propose a novel model named LSI-EGT that leverages the pre-trained contextualized MentalRoBERTa model alongside graph transformer.

III. METHODOLOGY

We focus on detecting stressors and significant stressful life events from mental health-related posts collected from various sources. Specifically, we have a dataset $\{X_i, Y_i\}$ where $i = 1, 2, \dots, N$, N is the dataset size, and each X_i is a text segment labelled with Y_i , representing the stressor associated with the text data. The task requires identification of specific stressors from the text data. This can be formalized as a multi-class text classification problem where the goal is to classify each text segment X_i into one of the predefined stressor categories $Y_i \in \{Y_1, Y_2, \dots, Y_C\}$, where C is the number of stressor categories. To achieve this, we develop a function with parameters θ to best classify each X_i into the correct stressor category

$$\arg \max_{\theta} \prod_{i=1}^N P(Y_i | X_i, \theta). \quad (1)$$

An overview of the proposed model is illustrated in Fig. 1 and the algorithm is provided in Algorithm 1. The process begins with (a) collecting and preprocessing the raw data. Then, (b) the MentalRoBERTa model is used to create embeddings for text. (c) These embeddings are used to build a similarity matrix using cosine similarity. A graph is then constructed where each node represents a text segment, and edges are based on cosine similarity between segments. Finally, (d) a graph neural network-based model is applied where, the final node representations are used to classify each text segment into stressor categories.

A. Text Preprocessing

To prepare texts for input into the MentalRoBERTa model, each text T_i from the dataset $\mathcal{D} = \{T_i\}_{i=1}^N$ undergoes preprocessing steps. First, each text T_i is tokenized using the MentalRoBERTa tokenizer, producing a sequence of tokens $\text{tokens}_i = [\text{token}_{i,1}, \text{token}_{i,2}, \dots, \text{token}_{i,|\text{tokens}_i|}]$, where $|\text{tokens}_i|$ denotes the number of tokens in T_i . Next, special tokens are added to each token sequence to denote the beginning and end of a sequence $\text{tokens}_{\text{special},i} = [[\text{CLS}], \text{token}_{i,1}, \dots, \text{token}_{i,|\text{tokens}_i|}, [\text{SEP}]]$. These tokens are then encoded into numerical IDs using the tokenizer's vocabulary, resulting in $\text{input_ids}_i = [\text{id}_{i,1}, \text{id}_{i,2}, \dots, \text{id}_{i,|\text{input_ids}_i|}]$, where $\text{id}_{i,j}$ represents the numerical ID of the j th token in T_i . If $|\text{input_ids}_i| < L$, the sequence is padded with a special padding token. If $|\text{input_ids}_i| > L$, it is truncated. Finally, these padded or truncated sequences $\{\text{input_ids}_{\text{padded},i}\}_{i=1}^N$ are organized into a tensor $\mathbf{X}$ of shape $N \times L$, where N is the number of texts in the dataset and L is the fixed length of each input sequence.

B. MentalRoBERTa Embeddings

For the purpose of generating embeddings for our text, MentalRoBERTa is used. MentalRoBERTa [37] is trained on mental health posts from social media and customized for the detection of mental health conditions. MentalRoBERTa, such as the original RoBERTa model, is based on a transformer architecture comprising multiple layers of self-attention mechanisms and feed-forward neural networks. Each layer processes the input sequence and produces refined embeddings that capture complex patterns and relationships. The input sequence is defined as $X_i = [[\text{CLS}], \text{token}_{i,1}, \dots, \text{token}_{i,|\text{tokens}_i|}, [\text{SEP}]]$. The $[\text{CLS}]$ token, added at the beginning of each input sequence, captures a summary representation of the entire sequence. Each transformer layer l transforms these embeddings

$$E^{(l+1)} = \text{TransformerLayer}(E^{(l)}) \quad (2)$$

where $E^{(0)} = \{E_{[\text{CLS}]}, E_1, E_2, \dots, E_L\}$. We used the final hidden state of the $[\text{CLS}]$ token from the last layer L to generate the embeddings for the input text

$$\text{Embedding}(X_i) = H_{[\text{CLS}]}^{(L)} \quad (3)$$

Algorithm 1: Stressor Identification Using LSI-EGT.**Require:** Texts from SAD, and SMOTE**Ensure:** Classification of texts into stressor categories

- 1: Read datasets (SAD, SMOTE).
- 2: Obtain [CLS] token embeddings using MentalRoBERTa:

$$embedding_i = \text{MentalRoBERTa}(U_i) \in \mathbb{R}^{N \times D_h} \quad (16)$$

- 3: Construct similarity matrix (*sim_matrix*) using cosine similarity between sentence embeddings:

$$sim_matrix_{ij} = \cos(embedding_i, embedding_j) \quad (17)$$

- 4: Create nodes for each text instance with *embedding* as node features.

$$nfeat_i = embedding_i \quad (18)$$

- 5: Define edges based on top k most similar neighbors from the similarity matrix:

$$adj_matrix_{ij} = \begin{cases} 1 & \text{if } j \in \text{top } k \text{ similar neighbor of } i \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

- 6: Create Train and Test mask as *train_mask* and *test_mask* respectively.

- 7: Define the model class

$$model = \text{GAT} \left(\begin{matrix} input_feat = nfeat[0].shape[1], \\ num_classes \end{matrix} \right) \quad (20)$$

- 8: Initialize the Optimizer (*Opt*) and Loss Function (*Loss*):

$$Opt = \text{Adam}(model.parameters(), lr = 0.001) \quad (21)$$

$$Loss = \text{CrossEntropyLoss}() \quad (22)$$

- 9: For each Training epoch

- I. *model.train()*
- II. *Opt.zero_grad()*
- III. *predictions = model(nfeat, adj_matrix)*
- IV. *loss = Loss(pred[train_mask], target[train_mask])*
- V. *loss.backward()*
- VI. *Opt.step()*

- 10: Model Evaluation

- I. *model.eval()*
- II. *pred=model(nfeat,adj_matrix)*
- III. *test_acc=accuracy(pred[test_mask], target[test_mask])*

- 11: Return the trained model *model*.

C. Graph Construction

Given the embeddings as $\{X_0, X_1, \dots, X_N\}$, where X_i represents the embedding of the i th instance. The similarity matrix, *sim_matrix* of size $N \times N$ is created based on the cosine similarity between the text embeddings, where N is the size of the dataset. Cosine similarity measures the cosine of the angle between two vectors, providing a value between -1 and 1 . For our purpose, it indicates how similar two pieces of text are.

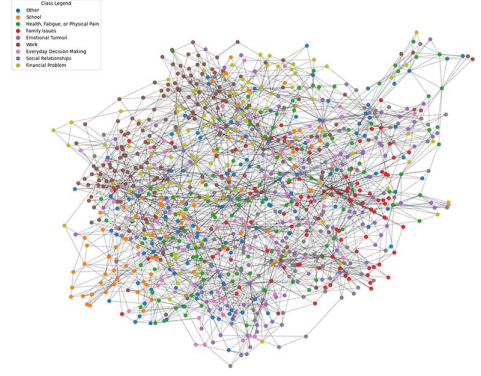


Fig. 2. Generated graph for the SAD dataset using 1000 nodes with $k = 5$.

The similarity matrix, provides a pairwise similarity score for all texts in our dataset. Mathematically, the cosine similarity between two text embeddings $\mathbf{X}_i$ and $\mathbf{X}_j$ is defined as:

$$sim_matrix_{ij} = \text{cosine_similarity}(\mathbf{X}_i, \mathbf{X}_j) = \frac{\mathbf{X}_i \cdot \mathbf{X}_j}{|\mathbf{X}_i| \cdot |\mathbf{X}_j|}. \quad (4)$$

Here, $\mathbf{X}_i \cdot \mathbf{X}_j$ represents the dot product of $\mathbf{X}_i$ and $\mathbf{X}_j$, and $|\mathbf{X}_i|$ and $|\mathbf{X}_j|$ denote the Euclidean norms of $\mathbf{X}_i$ and $\mathbf{X}_j$, respectively.

Let $G = (V, E)$ be a graph with nodes V and edges E . $V = \{v_1, v_2, \dots, v_N\}$ represent the N nodes in the graph. The nodes are described by $X \in \mathbb{R}^{N \times M}$ where M is the size of embeddings. Nodes are partitioned into training set V_{train} and test set V_{test} , such that $V_{\text{train}} \cup V_{\text{test}} = V$ and $V_{\text{train}} \cap V_{\text{test}} = \emptyset$. For each node v_i , we identified the top k most similar neighbors based on the similarity scores in *sim_matrix*. The value of k is chosen from the range [7], [12]. Node v_i is connected with its top k neighbors by adding edges. Let $N_k(v_i)$ denote the set of top k neighbors of node v_i . The edge set E is defined as

$$E = \{(v_i, v_j) \mid v_j \in N_k(v_i) \text{ for } v_i \in V\}. \quad (5)$$

A sample graph generated for the stress annotated dataset (SAD) dataset is shown in Fig. 2.

D. Classification With Graph Transformer

In this study, we utilize graph transformer instead of a fully connected transformer to better model the relationships between stressors in mental health discourse. While fully connected transformers employ self-attention over all tokens, they do not inherently capture structured dependencies between stressors. This limitation is particularly relevant in stress classification, where certain stressors, such as “family issues” and “social isolation” have strong interdependencies. Graph Transformers address this challenge by leveraging a structured similarity graph, where edges represent meaningful relationships between stressors. By incorporating graph-based attention, our approach allows information propagation between related stressors.

The graph transformer [38] model extends the graph attention network (GAT) by employing averaging for multihead outputs. For the proposed each layer in the model includes Graph

Transformer Layer, LayerNormalization, ReLU with the output of i th layer represented by $\{h_0^{(l)}, h_1^{(l)}, h_2^{(l)}, \dots, h_N^{(l)}\}$. For the graph transformer layer, given an edge from node i to j , for c_{th} head at layer l , query $q_{c,i}^{(l)}$ and key vectors $k_{c,i}^{(l)}$ are calculated for node i and j respectively as follows:

$$q_{c,i}^{(l)} = W_{c,q}^{(l)} h_i^{(l)} + b_{c,q}^{(l)} \quad (6)$$

$$k_{c,j}^{(l)} = W_{c,k}^{(l)} h_j^{(l)} + b_{c,k}^{(l)}. \quad (7)$$

Then, attention coefficients $\alpha_{c,ij}^{(l)}$ are then calculated as

$$\alpha_{c,ij}^{(l)} = \frac{\langle q_{c,i}^{(l)}, k_{c,j}^{(l)} \rangle}{\sum_{u \in N(i)} \langle q_{c,i}^{(l)}, k_{c,u}^{(l)} \rangle} \quad (8)$$

where $\langle q, k \rangle = \exp\left(\frac{q^T k}{\sqrt{d}}\right)$. After obtaining the multi-head attention from the graph, message aggregation is performed from the distant node j to the source node i

$$v_{c,j}^{(l)} = W_{c,v}^{(l)} h_j^{(l)} + b_{c,v}^{(l)} \quad (9)$$

$$h_i^{(l+1)} = \parallel_{c=1}^C \left[\sum_{j \in N(i)} \alpha_{c,ij}^{(l)} (v_{c,j}^{(l)}) \right]. \quad (10)$$

Further, gated residual connections are incorporated to prevent over-smoothing and enhance model expressiveness by electively combines the input and output of a graph transformer layer. This mechanism allows the model to decide how much of the new information (output of the layer) should replace the old information (input to the layer), thereby controlling the flow of information

$$r_i^{(l)} = W_r^{(l)} h_i^{(l)} + b_r^{(l)} \quad (11)$$

$$\beta_i^{(l)} = \text{sigmoid} \left(W_g^{(l)} \left[\widehat{h_i^{(l+1)}}; r_i^{(l)}; \widehat{h_i^{(l+1)}} - r_i^{(l)} \right] \right). \quad (12)$$

Then layer normalization and ReLU are added. LayerNormalization ensure that each node's activation has zero mean and unit variance and ReLU introduces nonlinearity into the graph transformer's layer, allowing it to learn complex mappings between the input and output embeddings

$$h_i^{(l+1)} = \text{ReLU} \left(\text{LayerNorm} \left((1 - \beta_i^{(l)}) \widehat{h_i^{(l+1)}} + \beta_i^{(l)} r_i^{(l)} \right) \right). \quad (13)$$

For the final layer output, final node representations $h_i^{(l+1)}$ are calculated by averaging multihead outputs and applying linear transformations

$$\widehat{h_i^{(l+1)}} = \frac{1}{C} \sum_{c=1}^C \sum_{j \in N(i)} \alpha_{c,ij}^{(l)} (v_{c,j}^{(l)} + e_{c,ij}^{(l)}) \quad (14)$$

$$h_i^{(l+1)} = (1 - \beta_i^{(l)}) \widehat{h_i^{(l+1)}} + \beta_i^{(l)} r_i^{(l)}. \quad (15)$$

IV. EXPERIMENTAL SETUP AND RESULTS

This section presents the experimental setup, including dataset details, implementation specifics, and evaluation metrics.

A. Datasets

The proposed model is evaluated on two datasets: SAD and a synthetically generated dataset named as SMOTE. The statistical details of each dataset are listed in Table II.

1) *SAD Dataset* [39]: A human-labeled stress factor detection dataset. The stressor set includes nine stressor categories: school, financial problem, family issues, social relationships, work, health, fatigue, or physical pain, emotional turmoil, other, everyday decision making. The dataset contains 6850 SMS-like sentences, where each post is a short message (normally one sentence) divided into one of these categories. Specifically, we focused on instances labeled as positive, resulting in a subset of 6475 instances.

2) *SMOTE Dataset*: The SMOTE dataset contains 6128 messages. The dataset was created using LLM models, with 32 classes. Some examples from the dataset are provided in Table III. A total of eight most common stressors were identified. These stressors were further classified into 32 subcategories

- 1) *Health Issues*: Physical, medication, mental.
- 2) *Academic Distress*: Substance abuse, exam pressure/failure, social isolation.
- 3) *Relationship Issues*: Parent-child, spouse/ fiancé/ boyfriend/ girlfriend, domestic violence, friction with in-laws.
- 4) *Loss*: Death and loss, breakup/divorce.
- 5) *Financial Issues*: Loss in property, loss in business, job loss, income-expenditure imbalance, loan/debt.
- 6) *Female Related Issues*: Infertility, hormonal imbalance, premenstrual syndrome, menopause, postpartum depression.
- 7) *Being Bullied*: Religion, racial cultural bullying, psychological/emotional bullying, social bullying, verbal bullying, physically bullied, gender, bullying on age.
- 8) *Others*: Trauma, body image issues.

B. Model Summary

The proposed model was benchmarked against several established models:

BERT: Initialized from pre-trained BERT-base weights, utilized the [CLS] token embedding for decoding, followed by a Dense Layer for classification.

RoBERTa and MentalRoBERTa: Both RoBERTa and MentalRoBERTa utilized the [CLS] token embedding for decoding, followed by a Dense Layer for classification.

CNN: For the CNN model, FastText embeddings were employed with a three-channel convolution as the encoder. This setup featured 128-dimensional embeddings and filters of sizes 3, 4, and 5, each with 100 filters per size. Postlevel embeddings were obtained via max-pooling, and concatenated features from different filters were used for decoding.

Further, graph-based models GraphConv [40], SAGEConv [41], graph attention network (GATConv) [42], and transformer layer were integrated with these base models. BERT, RoBERTa, and MentalRoBERTa [CLS] tokens and SBERT embeddings were used as features to create the graph. For the SBERT model, "all-mpnet-base-v2" was used for the SMOTE Dataset, and the "all-MiniLM-L12-v2" model was used for the SAD

TABLE II
DATASETS DESCRIPTION

Dataset	Data Source	Total Size	Size Used	Average Tokens per Instance	Train Size	Test Size
SAD	SMS	6850	6475	63	5180	1295
SMOTE	LLM	6128	6128	70	4902	1226

TABLE III
SOME ILLUSTRATIONS FROM THE SMOTE (PROPOSED DATASET)

Text	Label
Intrusive thoughts invade my mind, tormenting me with their relentless persistence.	Mental
The fear of disappointing my loved ones through academic failure fuels my depressive emotions.	Exam Stress/Failure
I'm suffocating under the weight of their emotional distance, feeling like a stranger in my own relationship and sinking into depression.	Spouse/Fiancé/Boyfriend/Girlfriend
The divorce has shattered my sense of security, leaving me to rebuild my life on a foundation that feels fragile and uncertain.	Divorce/Breakup
The constant job rejections make me question my career choices, wondering if I should pursue a different path.	Job Loss
I struggle with a lack of focus and concentration during PMS, making it hard to stay productive.	PMS
The constant judgment and prejudice directed at my gender rob me of my joy and confidence.	Gender
I'm tired of feeling like my worth is defined by my physical appearance, rather than my inherent value as a human being.	Body Image Issues

TABLE IV
MODEL PERFORMANCE ANALYSIS FOR SAD AND SMOTE DATASETS

	SAD				SMOTE			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
BERT	63.16	62.09	63.23	62.40	92.41	92.50	91.69	91.89
Roberta	65.86	64.81	65.82	64.94	94.61	94.87	93.72	94.29
MentalRoBERTa	64.94	63.77	64.75	64.06	94.61	94.67	93.80	93.97
CNN	64.75	64.74	64.78	64.42	94.64	94.92	94.53	94.59
BERT+GraphConv	51.11	50.71	50.50	50.19	84.58	84.14	83.50	83.65
BERT+GraphSAGE	61.62	61.10	61.10	60.74	92.82	92.56	92.51	92.40
BERT+GATConv	46.48	46.42	46.25	46.22	82.21	82.52	81.31	81.60
BERT+TransformerConv	63.47	63.40	63.20	63.21	93.39	93.22	93.00	92.95
SBERT+GraphConv	64.01	63.85	63.22	63.19	95.26	95.55	94.87	95.01
SBERT+GraphSAGE	66.48	65.68	66.19	65.83	96.81	96.85	96.42	96.55
SBERT+GATConv	64.78	64.76	64.07	64.16	94.45	94.45	93.94	94.06
SBERT+TransformerConv	68.18	67.32	67.55	67.09	96.90	96.93	96.51	96.61
Roberta+GraphConv	58.53	57.59	57.33	57.14	93.38	93.71	93.43	93.49
Roberta+SAGEConv	65.48	64.67	64.84	64.43	96.65	96.62	96.44	96.49
Roberta+GATConv	53.28	52.55	52.58	52.25	91.92	91.89	91.43	91.48
Roberta+TransformerConv	66.94	66.27	66.13	65.89	96.90	96.85	96.64	96.71
MentalRoBERTa+GraphConv	62.47	60.81	61.35	60.54	93.71	93.78	93.53	93.48
MentalRoBERTa+SAGEConv	68.33	67.63	67.61	67.38	96.73	96.78	96.65	96.65
MentalRoBERTa+GATConv	61.23	60.36	60.49	60.24	92.08	91.96	91.56	91.66
LSI-EGT	68.73	68.26	68.03	67.93	97.38	97.36	97.21	97.26

Note: Bold indicates the final total results.

dataset. *BERT/RoBERTa/MentalRoBERTa/SBERT+GraphConv*: BERT/RoBERTa/MentalRoBERTa [CLS] token embeddings, or SBERT sentence embeddings are used to create graph and used as node features. GraphConv is then used for classification.

BERT/RoBERTa/MentalRoBERTa/SBERT+SAGEConv: BERT/RoBERTa/MentalRoBERTa [CLS] token embeddings, or SBERT sentence embeddings are used to create graph and used as node features. SAGEConv is used for classification.

BERT/RoBERTa/MentalRoBERTa/SBERT+GATConv: BERT/RoBERTa/MentalRoBERTa [CLS] token embeddings, or SBERT sentence embeddings are used to create graph and used as node features. GATConv is used for classification.

BERT/RoBERTa/MentalRoBERTa/SBERT+Transformer Layer: BERT/RoBERTa /MentalRoBERTa [CLS] token

embeddings, or SBERT sentence embeddings are used to create graph and used as node features. Transformer layer is then used for classification.

C. Experiment Configuration

All experiments were conducted on Google Colab with 16 GB of memory. For hyper-parameter settings, hidden dimension size, D_h was set to 256. The framework and initial weights of the MentalRoBERTa were sourced from Huggingface's Transformers. The Pytorch Geometric framework was used for Graph Transformer implementation. The AdamW optimizer was employed for model training, with a batch size N_b of 16 on the SAD, and SMOTE Datasets. The value of k (top k neighbors) was chosen to be 7. A learning rate of $1e-5$ was used, and a dropout rate of 0.3 was set for all experiments. Experimental

TABLE V
LSI-EGT F1-SCORE ACROSS DIFFERENT TRAINING DATA SIZES FOR SAD AND SMOTE DATASETS

Model Name	SAD								SMOTE							
	20%	30%	40%	50%	60%	70%	80%		20%	30%	40%	50%	60%	70%	80%	
MentalRoBERTa + GraphConv	52.80	55.94	57.52	59.12	60.41	60.24	62.31		85.21	88.15	88.89	89.67	90.83	90.65	93.48	
MentalRoBERTa + SAGEConv	60.15	61.89	63.40	64.53	65.58	67.12	67.38		89.32	91.52	92.67	93.35	94.29	94.63	96.65	
MentalRoBERTa + GATConv	52.25	53.24	55.18	56.74	57.25	58.48	60.26		84.46	86.91	88.56	89.21	90.97	90.70	91.66	
Proposed Method-LSI-EGT	58.86	58.59	64.53	65.21	65.91	67.67	67.93		89.00	92.22	92.62	94.75	95.14	95.45	97.26	

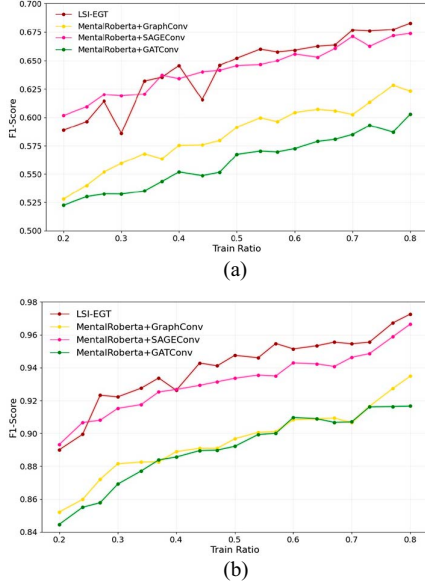


Fig. 3. F1-score results on: (a) SAD; and (b) SMOTE under different training data proportions.

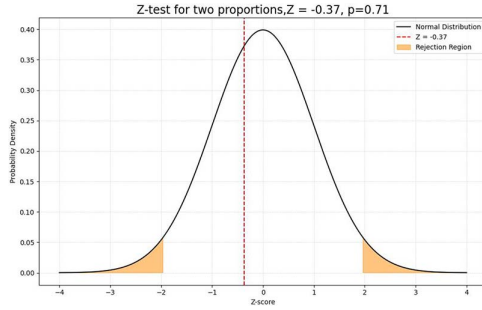


Fig. 4. Graphical representation of the critical region of the hypothesis.

configurations were specified as `configs = {'num_layers': 4, 'hidden_dim': 256, 'num_heads': 2, 'dropout': 0.3, 'lr': 0.001, 'epochs': 500, 'weight_decay': 0.0005}`. The evaluation metrics chosen were accuracy, precision, recall, and F1-score.

D. Results and Analysis

The overall experiment results of our model and the baselines are listed in this section, with results described in Tables IV and V. The class-wise performance analysis for both datasets is provided in Tables VI and VII. Fig. 3 illustrates the visualizations of the results obtained with different sizes of training data. Fig. 4 shows the graphical representation of the critical region.

TABLE VI
CLASS-WISE PERFORMANCE ANALYSIS FOR THE SAD DATASET

Class	Total	Train	Test	Precision	Recall	F1-Score
School	641	513	128	78.57	85.94	82.09
Financial Problem	780	624	156	73.42	74.36	73.89
Family Issues	606	485	121	68.64	66.94	67.78
Social Relationships	539	431	108	63.83	55.56	59.41
Work	968	774	194	74.40	79.38	76.81
Health, fatigue, or physical pain	826	661	165	69.19	77.58	73.14
Emotional Turmoil	755	604	151	69.57	52.98	60.15
Other	786	629	157	55.35	56.05	55.70
Everyday Decision Making	574	459	115	61.34	63.48	62.39
Total - Micro Average	6475	5180	1295	68.73	68.73	68.73
Total - Macro Average	6475	5180	1295	68.26	68.03	67.93
Total - Weighted Average	6475	5180	1295	68.58	68.73	68.43

TABLE VII
CLASS-WISE PERFORMANCE ANALYSIS FOR THE SMOTE DATASET

Class	Total	Train	Test	Precision	Recall	F1-Score
Infertility	100	80	20	95.70	92.00	93.80
Death and Loss	150	120	30	94.00	85.00	89.30
Physical	230	184	46	98.00	98.00	98.00
Parent-Child	200	160	40	82.00	89.00	85.40
Verbal Bullying	200	160	40	90.00	98.00	93.80
Trauma	199	159	40	100.00	98.50	99.20
Loan/Debt	150	120	30	93.00	82.00	87.20
Income-Expenditure Imbalance	150	120	30	80.00	92.00	85.60
Bullying on Age	180	144	36	93.00	100.00	96.40
Friction with in-laws	160	128	32	92.00	100.00	95.80
Hormonal Imbalance	300	240	60	97.00	97.00	97.00
Mental	170	136	34	95.00	97.50	96.20
Social Bullying	200	160	40	95.00	89.00	91.90
Menopause	200	160	40	96.00	98.00	97.00
Failure/Exam Pressure	300	240	60	100.00	100.00	100.00
Gender	234	187	47	100.00	96.50	98.20
Spouse/Fiance/Boyfriend/Girlfriend	150	120	30	97.00	85.00	90.60
Body Image Issues	200	160	40	100.00	100.00	100.00
Postpartum Depression	200	160	40	100.00	95.50	97.70
Medication	200	160	40	96.00	100.00	97.90
Racial Cultural Bullying	200	160	40	100.00	100.00	100.00
Religion	200	160	40	100.00	100.00	100.00
Job Loss	180	144	36	97.50	97.50	97.50
Loss in business	150	120	30	100.00	91.00	95.30
Loss in property	150	120	30	100.00	97.00	98.50
Substance abuse	114	91	23	100.00	100.00	100.00
Breakup/divorce	348	278	70	95.00	97.50	96.20
Domestic violence	165	132	33	97.50	100.00	98.70
Psychological Emotional Bullying	200	160	40	97.50	93.50	95.40
Premenstrual Syndrome	198	159	39	100.00	100.00	100.00
Social Isolation	200	160	40	100.00	100.00	100.00
Physically Bullied	150	120	30	100.00	100.00	100.00
Total - Micro Average	6128	4902	1226	97.38	97.38	97.39
Total - Macro Average	6128	4902	1226	97.36	97.21	97.26
Total - Weighted Average	6128	4902	1226	97.51	97.38	97.39

We first compared our proposed model with state-of-the-art baseline models on the SAD Dataset. The BERT model achieved an F1-score of 62.4%, while RoBERTa performed slightly better with an F1-score of 64.94%. MentalRoBERTa achieved an F1-score of 64.06%, outperforming BERT. The CNN model attained an F1-score of 64.42%. Next, we integrated graph convolution layers with BERT, SBERT, RoBERTa, and MentalRoBERTa and evaluated their performances. BERT with GraphConv yielded an F1-score of 50.19%, 60.74% with SAGEConv, 46.22% with GATConv and 63.21% with Graph Transformer. SBERT outperformed with GraphConv yielded an F1-score of 63.19%, 65.83% with SAGEConv, 64.16% with

GATConv, and 67.09% with Graph Transformer. RoBERTa performed better than BERT, achieving an F1-score of 57.14% with GraphConv, 64.43% with SAGEConv, 52.25% with GAT, and 65.89% with graph transformer. MentalRoBERTa outperformed all other models, achieving an F1-score of 60.54% with GraphConv, 67.38% with SAGEConv, 60.24% with GATConv, and 67.93% with Transformer Layer. The performance of various models was evaluated across training ratios ranging from 20% to 80%. The models included combinations of MentalRoBERTa with GraphConv, SAGEConv, and GATConv, as well as a proposed method labeled LSI-EGT. It can be observed that F1-Score for almost all of the methods gets better while the proportions of training data increase. When 70% is reached by the proportion, the ascending tendency of performance tends to be gentle. The proposed method (LSI-EGT) generally outperformed the other models. MentalRoBERTa + SAGEConv achieved second-best with reaching 67.38% F1-Score at 80% training ratio.

We also evaluated the performance of the models on the SMOTE dataset. BERT achieved an F1-score of 91.89, while RoBERTa performed better with an F1-score of 94.29. MentalRoBERTa had an F1-score of 93.97, and CNN achieved an F1-score of 94.59. When incorporating graph convolution layers, BERT + GraphConv yielded an F1-score of 83.65, BERT+GraphSAGE achieved 92.40, BERT+GAT scored 81.60, and BERT + Transf reached 92.95. SBERT with graph neutral network layers showed F1-scores of 95.01 for SBERT + GraphConv, 96.55 for SBERT + GraphSAGE, 94.06 for SBERT + GAT, and 96.61 for SBERT + Transf. RoBERTa with GCNConv achieved the F1-score of 93.49, 96.49 for RoBERTa+SAGEConv, 91.48 for RoBERTa + GAT, and 96.71 for RoBERTa + TransformerLayer. MentalRoBERTa with graph convolution layers showed F1-scores of 93.48 for MentalRoBERTa + GraphConv, 96.65 for MentalRoBERTa + SAGEConv, 91.66 for MentalRoBERTa+ GAT, and 97.26 for MentalRoBERTa + Transformer. Further the performance was also evaluated across training ratios ranging from 20% to 80% using combinations of MentalRoBERTa with GraphConv, SAGEConv, and GATConv, and the proposed method LSI-EGT. F1-Scores for almost all of the methods get better while the proportions of training data increase. LSI-EGT demonstrates the highest overall performance and the most significant gains with increased training data, while MentalRoBERTa + SAGEConv achieved second-best.

E. OOD Testing

To evaluate the model's robustness, we conducted out-of-distribution (OOD) testing by introducing stress-related text samples from categories not present in the training data. Table VIII presents these samples along with their actual labels, predicted probability, and predicted labels. Several misclassifications indicate model biases, such as associating *social media stress* with *body image issues* and *environmental anxiety* with *business loss* due to lexical similarities. The model identified *survivor's guilt* related stress as *trauma* with high confidence (0.9056) and exhibited uncertainty (0.1368) in

predicting climate-related stress. Additionally, executive burnout was misclassified as *business loss*, reflecting an overlap between professional and financial stress. The low confidence in some predictions suggests ambiguity in recognizing novel stressors. These findings highlight the need for more diverse training data, improved handling of uncertainty. Future work should explore contrastive learning and domain adaptation techniques to mitigate misclassification errors and refine stress detection in real-world scenarios.

F. Time Complexity Analysis

In this section, asymptotic time complexity of the proposed approach has been assessed.

- 1) *Text Preprocessing*: Tokenizing the input text for all samples takes time proportional to the total number of tokens in the dataset. If l is the average number of tokens per sample, the complexity is $O(n \cdot l)$. Padding and truncating all samples to a fixed length is linear in the number of samples: $O(n)$. Generating attention masks for all samples is $O(n \cdot l)$.
- 2) *Generating Embeddings*: The forward pass involves the attention mechanism, which has a complexity of $O(l \cdot m^2)$ per sample, leading to $O(n \cdot l \cdot m^2)$ for the entire dataset, where m is the hidden size of the model.
- 3) *Similarity Matrix Generation*: The time complexity for calculating pairwise cosine similarity given n embeddings of dimension d is $O(n^2 d)$.
- 4) *Graph Construction*: The time complexity for finding the top k most similar embeddings based on the cosine similarity matrix for each instance is: $O(n^2 \log k)$. The time complexity for creating a graph where each instance in the dataset represents a node and the top k similar nodes are neighbors of that node is: $O(n \cdot k)$.
- 5) *Graph Transformer*: The attention mechanism, in the graph transformer architecture, has a complexity of $O(n^2)$, where n is the number of nodes in the graph.

G. Validation of Proposed Approach Through Statistical Approach Through Statistical Significance

Z-test was used to validate the performance of the LSI-EGT model. Utilizing statistical significance, researchers validate their conclusions by testing the validity of their models. Using the dataset of SMOTE, two samples were picked with varying sizes. In first sample, 450(n_1) reviews were considered out of which 437(x_1) were correctly classified with proportion(p_1)=0.9711. In second 750(n_2) reviews were considered out of which 731(x_2) were correctly classified with proportion (p_2)=0.9746. To test the validation of the proposed model, the null hypothesis(H_0) and alternative hypothesis(H_a) are stated below:

Null Hypothesis (H_0): Both datasets have the same proportion of correctly labeled reviews, (p_1) = (p_2)

Alternative Hypothesis (H_a): Both datasets have a different proportion of correctly labeled reviews, (p_1) $\neq$ (p_2)

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}. \quad (23)$$

TABLE VIII
OUT-OF-DISTRIBUTION (OOD) TESTING RESULTS

Text	Actual Label	Predicted Probability	Predicted Label
Every notification feels like an expectation I can't meet. The pressure to appear perfect online is suffocating, making me question my self-worth.	Social media stress	0.2953	Body image issues
I can't shake off the anxiety when I see reports of climate disasters. My heart clenches knowing my existence contributes to the destruction of the planet.	Environmental anxiety	0.1368	Business loss
I walked away from the accident unharmed, but my best friend didn't. The 'why me?' haunts me more than the crash itself.	Survivor's guilt	0.9056	Trauma
I feel like I am trapped in my daily routine and have lost motivation.	Existential crisis	0.1959	Income expenditure imbalance
As a CEO, I can't show weakness. The loneliness of making tough decisions in silence gnaws at me, yet I must wear confidence like armor.	Executive burnout	0.8335	Business loss

TABLE IX
ERROR ANALYSIS: ACTUAL VERSUS PREDICTED LABELS

Text	Actual Label	Primary Prediction	Secondary Prediction
I feel like I'm drowning in their expectations, unable to find my own path amidst the weight of their judgment.	Parent-Child	Age	Parent-Child
I'm consumed by a sense of emptiness, as the void left by their departure seems impossible to fill.	Death and Loss	Breakup	Death and Loss
Every hurtful remark and exclusionary act reinforce the feeling of being an outsider, as if I don't belong.	Social Bullying	Psychological/Emotional Bullying	Social Bullying
The hormonal imbalances of PCOS can disrupt my sleep patterns, leading to insomnia or restless nights that further exacerbate my fatigue and mood.	PCOS	PCOS	Menopause

The value of pooled proportion $\hat{p}$ calculated using (24) is 0.9733: The Z-test for two population proportion using two tailed test was used. Equation (25) shows the calculation of Z-statistic

$$Z = \frac{(p_1 - p_2)}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = -0.37. \quad (24)$$

Thus, the null hypothesis is not rejected at the 0.05 significance level. This implies that the accuracy of the proposed model is consistent throughout different samples of the same dataset. Thus the result of the proposed model can be trusted.

H. Error Analysis

To understand the model's misclassifications, an error analysis was conducted on cases where the predicted label differed from the ground truth. For each misclassified instance, the top two predicted classes—the class with the highest probability and the second most likely class—were examined. Results are added in Table IX. For example, a statement about feeling overwhelmed by parental expectations was predicted as age but the correct label was parent child. A sentence about feeling empty after someone's departure was predicted as breakup instead of death and loss though it was correctly second-labeled as death and loss. Another example was a description of bullying, which was predicted as psychological emotional bullying instead of the correct label social bullying. A statement about PCOS affecting sleep was correctly predicted as PCOS but also

secondary predicted as menopause. These findings indicate that while the model is able to identify relevant stressor categories, it sometimes struggles with overlapping features between similar stressors. Future improvements could involve multilabel classification strategies, or hierarchical classification models to better capture these distinctions.

V. CONCLUSION AND FUTURE WORK

The proposed model LSI-EGT represents a significant advancement in the field of automated stressors detection, exhibiting superior performance over traditional models across various datasets. However, the model's limitations, particularly in detecting stressors in SAD dataset, highlight areas for further improvement. Future research should focus on enhancing the model's sensitivity to these nuanced stressors and addressing data imbalance issues to improve its overall reliability. A key challenge in the proposed model is the computational cost of generating a similarity matrix and constructing a graph. The current approach builds a single unified graph for the dataset, which is feasible given the dataset's size. However, for larger datasets, the method scales quadratically with the number of samples, posing constraints in real-time settings. To mitigate this, several strategies can be considered: Subgraph construction, approximate nearest neighbor (ANN) search, graph sampling methods. One possible way to enhance stressor classification is by incorporating large language models (LLMs) such as GPT, and LLaMA. These models are trained on diverse datasets, enabling them to capture deeper contextual relationships and improve classification accuracy. Unlike traditional transformer models, LLMs can generalize better to unseen stressors and require less labeled data due to their few-shot and zero-shot learning capabilities. However, integrating LLMs presents challenges, such as high computational requirements and increased memory usage. Additionally, their black-box nature makes it difficult to interpret how they establish relationships between stressors. Future research could explore hybrid models that combine LLMs with graph-based approaches to leverage both rich contextual understanding and structured relationship modeling for more effective stressor detection.

REFERENCES

- [1] I. Kupcova, L. Danisovic, M. Klein, and S. Harsanyi, "Effects of the COVID-19 pandemic on mental health, anxiety, and depression," *BMC Psychol.*, vol. 11, no. 1, p. 108, Apr. 2023.
- [2] N. Salari et al., "Prevalence of stress, anxiety, depression among the general population during the COVID-19 pandemic: a systematic review and meta-analysis," *Globalization Health*, vol. 16, no. 1, p. 57, Jul. 2020.

- [3] R. Meena and R. Paliwal, "Impact of social media on adolescent's stress," *Bengal, Past Present: J. Calcutta Historical Soc.*, vol. 117, pp. 372–378, 2022.
- [4] "Gen Z more likely to report mental health concerns", APA. 2023. [Online]. Available: <https://www.apa.org/monitor/2019/01/gen-z>
- [5] L. Kauhanen et al., "A systematic review of the mental health changes of children and young people before and during the COVID-19 pandemic," *Eur. Child Adolescent Psychiatry*, vol. 32, no. 6, pp. 995–1013, 2023.
- [6] J. Twenge, A. B. Cooper, T. Joiner, M. Duffy, and S. Binau, "Age, period, and cohort trends in mood disorder indicators and suicide-related outcomes in a nationally representative dataset, 2005-2017," *J. Abnormal Psychol.*, vol. 128, no. 3, p. 185, 2019.
- [7] American Psychological Association. "Stress in America 2020: A National Mental Health Crisis," 2020. [Online]. Available: <https://www.apa.org/news/press/releases/stress/2020/report-october>
- [8] National Academies of Sciences, Engineering, and Medicine, "Social Isolation and Loneliness in Older Adults: Opportunities for the Health Care System," Washington, DC: The Nat. Academies Press, 2020. doi: 10.17226/25663.
- [9] American Psychological Association, "Too much coronavirus media exposure may be bad for your health," 2022. [Online]. Available: <https://www.apa.org/pubs/highlights/spotlight/issue-181>
- [10] N. G. Swanson, "Working women and stress," *J. Amer. Med. Women's Assoc.* (1972), vol. 55, no. 2, pp. 76–79, 2000.
- [11] M. B. Berryhill and J. A. Durtschi, "Understanding Single Mothers' Parenting Stress Trajectories," *Marriage Family Rev.*, vol. 53, no. 3, pp. 227–245, 2016.
- [12] M. S. Epifanio, V. Genna, C. De Luca, M. Roccella, and S. La Grutta, "Paternal and maternal transition to parenthood: the risk of postpartum depression and parenting stress," *Pediatric Rep.*, vol. 7, no. 2, p. 5872, 2015.
- [13] O. Papalou and E. Diamanti-Kandarakis, "The role of stress in PCOS," *Expert Rev. Endocrinol. Metabolism*, vol. 12, no. 1, pp. 87–95, 2016.
- [14] D. Bhargava and H. Trivedi, "A study of causes of stress and stress management among youth," *IRA-Int. J. Manage. Social Sci.*, vol. 11, no. 3, pp. 108–117, 2018.
- [15] J. Bienertova-Vasku, P. Lenart, and M. Scheringer, "Eustress and distress: neither good nor bad, but rather the same?" *Bioessays*, vol. 42, no. 7, 2020, Art. no. e1900238.
- [16] T. H. Holmes and R. H. Rahe, "The social readjustment rating scale," *J. Psychosomatic Res.*, vol. 11, no. 2, pp. 213–8, 1967.
- [17] G. Singh et al., "Presumptive stressful life events scale (psles) - a new stressful life events scale for use in India," *Indian J. Psychiatry*, vol. 26, no. 2, pp. 107–14, 1984.
- [18] R. Cochrane and A. Robertson, "The life events inventory: A measure of the relative severity of psycho-social stressors," *J. Psychosomatic Res.*, vol. 17, no. 2, pp. 135–139, 1973.
- [19] American Psychological Association, "Increased need for mental health care strains capacity [Press release]," [Online]. Available: <https://www.apa.org/news/press/releases/2022/11/mental-health-care-strains>
- [20] Q. Li, Y. Xue, L. Zhao, J. Jia, and L. Feng, "Analyzing and identifying teens' stressful periods and stressor events from a microblog," *IEEE J. Biomed. Health Inform.*, vol. 21, no. 5, pp. 1434–1448, Sep. 2017.
- [21] J. Du et al., "Extracting psychiatric stressors for suicide from social media using deep learning," *BMC Med. Inform. Decis. Making*, vol. 18, no. Suppl. 2, p. 43, 2018.
- [22] Y. Zhang et al., "Psychiatric stressor recognition from clinical notes to reveal association with suicide," *Health Inform. J.*, vol. 25, no. 4, pp. 1846–1862, Dec. 2019.
- [23] Y. Leung and F. Khalvati, "Exploring COVID-19-related stressors: topic modeling study," *J. Med. Internet Res.*, vol. 24, no. 7, 2022, Art. no. e37142, <https://www.jmir.org/2022/7/e37142>
- [24] J. Sawalha, M. Yousefnezhad, Z. Shah, M. R. G. Brown, A. J. Greenshaw, and R. Greiner, "Detecting presence of PTSD using sentiment analysis from text data," *Front Psychiatry*, vol. 12, Feb. 2022, Art. no. 811392.
- [25] S. Han et al., "Classifying social determinants of health from unstructured electronic health records using deep learning-based natural language processing," *J Biomed Inform.*, vol. 127, p. 103984, 2022.
- [26] R. Richie, V. M. Ruiz, S. Han, L. Shi, and F. Tsui, "Extracting social determinants of health events with transformer-based multitask, multilabel named entity recognition," *J. Amer. Med. Inform. Assoc.*, vol. 30, no. 8, pp. 1379–1388, Aug. 2023.
- [27] S. Raza et al., "Discovering social determinants of health from case reports using natural language processing: algorithmic development and validation," *BMC Digit Health*, vol. 1, no. 1, p. 35, 2023.
- [28] M. P. Raveau et al., "Natural language processing analysis of the psychosocial stressors of mental health disorders during the pandemic," *NPJ Mental Health Res.*, vol. 2, no. 1, p. 17, 2023.
- [29] O. Iparraguirre-Villanueva, C. Paulino-Moreno, A. Epifanía-Huerta, and C. Torres-Ceclén, "Machine learning models to classify and predict depression in college students," *Int. J. Interact. Mob. Technol.*, vol. 18, no. 14, pp. 148–163, Aug. 2024.
- [30] M. Li, X. Sun, and M. Wang, "Detecting depression with heterogeneous graph neural network in clinical interview transcript," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 1, pp. 1315–1324, Feb. 2024.
- [31] M. Jain, S. Jain, A. Jain, and B. Garg, "CDME-GAT: context-aware depression detection using multiembedding and graph attention networks in social media text," *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 6, pp. 7212–7222, Dec. 2024, doi: 10.1109/TCSS.2024.3402689.
- [32] S. Monteith et al., "Big data are coming to psychiatry: a general introduction," *Int. J. Bipolar Disorder*, vol. 3, no. 1, p. 21, 2015.
- [33] F. Ringeval et al., "AVEC 2019 workshop and challenge: state-of-mind, detecting depression with AI, and cross-cultural affect recognition," in *Proc. 9th Int. on Audio/Vis. Emotion Challenge and Workshop*, 2019, pp. 3–12.
- [34] A. E. W. Johnson et al., "MIMIC-III, a freely accessible critical care database," *Sci. Data*, vol. 3, no. 1, 2016.
- [35] K. Lybarger, M. Yetisgen, and O. Uzuner, "The 2022 N2C2/UW shared task on extracting social determinants of health," *J. Am. Med. Inform. Assoc.*, vol. 30, no. 8, pp. 1367–1378, 2023.
- [36] Allen Institute for AI, "Science-parse," 2019. [Online]. Available: <https://github.com/allenai/science-parse>
- [37] S. Ji, T. Zhang, L. Ansari, J. Fu, P. Tiwari, and E. Cambria, "MentalBERT: Publicly available pretrained language models for mental healthcare," in *Proc. 13th Lang. Resour. Eval. Conf.*, Marseille, France: Eur. Lang. Resour. Assoc., 2022, pp. 7184–7190.
- [38] Y. Shi, H. Zhengjie, S. Feng, H. Zhong, W. Wang, and Y. Sun, "Masked label prediction: unified message passing model for semi-supervised classification," 2021, pp. 1548–1554, *arXiv:2009.03509*, doi: 10.24963/ijcai.2021/214.
- [39] M. Mauriello, E. Lincoln, G. Hon, D. Simon, D. Jurafsky, and P. Paredes, "SAD: A stress annotated dataset for recognizing everyday stressors in SMS-like conversational systems," in *Extended Abstr. 2021 CHI Conf. Human Factors Comput. Syst.*, 2021, pp. 1–7, doi: 10.1145/3411763.3451799.
- [40] C. Morris et al., "Weisfeiler and Leman go neural: Higher-order graph neural networks," *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, pp. 4602–4609, 2019.
- [41] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*, Red Hook, NY, USA: Curran Associates Inc., 2017, pp. 1025–1035.
- [42] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.



Minni Jain received the B.Tech. degree from GGSIPU, the M.Tech. degree from NSUT east campus (IS), and the Ph.D. degree from DTU (CSE). She currently serves as an Assistant Professor with Delhi Technological University, New Delhi, India. Her research interests include natural language processing, mental health care, social media analysis, and artificial intelligence. She has authored numerous research papers published in prestigious journals and conferences, including IEEE TRANSACTIONS, ACM Transactions, Springer, and Wiley.



Roopal Shakya is currently working toward the B.Tech Program in computer engineering from Delhi Technological University, Delhi, India.